

Abdoulaye SAYOUTI SOULEYMANE

AI Team Lead @ Understand Tech

<https://abdoulaye-sayouti.github.io/>



Over 4 years of experience in AI/LLM/RAG development and 3D data processing solutions. Specialized in impactful open-source AI solutions that enhance product quality and operational efficiency.

Work Experience

Understand Tech

► AI Team Lead

July. 2025 - Present
Frontignan, France

I lead the AI strategy and innovation roadmap at Understand Tech. We develop a best-in-class enterprise AI platform focused on efficiency, reliability, and security. I'm responsible for building and scaling the AI department in alignment with business objectives.

Aiway

► R&D Artificial Intelligence Engineer

Oct. 2022 - Jun. 2025
Aix-en-Provence, France

At Aiway, I led the development, training, and evaluation of 3D AI models for oral scan segmentation and optimization. These models are the cornerstone of **EASYMODEL** (www.ai4dental.com), a cutting-edge SaaS platform for orthodontic model creation, serving dentists and orthodontists worldwide.

- ❖ **Process Optimization:** Optimized data preparation, training, and evaluation workflows to improve model performance and meet client needs.
- ❖ **3D AI Models Development:** Implemented advanced 3D data annotation strategies and trained semantic segmentation models for various dental applications, optimizing accuracy and ensuring compliance with client specifications.

Enedis IA Lab

► R&D Artificial Intelligence Engineer

Jun. 2021 - Sept. 2022
Marseille, France

At Enedis AI Lab, I designed and implemented AI solutions for energy data analysis and natural language processing, automating critical processes in the energy sector.

- ❖ **Data Processing and Analysis:** Engineered custom Named Entity Recognition (NER) datasets and processed electrical load curve data for actionable insights.
- ❖ **Models Development:** Trained, optimized and deployed various NLP models, including text classification and NER, with a focus on zero-shot learning to reduce operational costs.

Infineon Technologies

► Data Analyst Intern

May - Sept. 2020
Padova, Italy

- ❖ **Dashboard Development:** Designed and developed a dashboard for real-time visualization and validation of the company's software, firmware and hardware development processes, in compliance with ISO 26262 "Road vehicles - Functional safety" and Automotive SPICE standards.

Projets *(Details in my online portfolio)*

- ❖ **Secure RAG System Development:** Developed an open-source RAG pipeline combining hybrid encoding and retrieval with advanced reranking, winning **1st place** in UnderstandTech's international competition on [Trustii.io](https://trustii.io) platform ([Github link](#)). The solution enables secure offline AI document processing, featuring multi-format document handling and optimized GPU performance.
- ❖ **AI Fashion Innovation:** Trained a specialized AI model for African fashion, under the Inclusive Fashion AI (InFashAI) initiative. Fine-tuned Stable Diffusion v1.4 with over 100,000 images, most through web scraping to boost African fashion representation.

Education

M.Sc in Artificial Intelligence, Avignon University

► Center for Education and Research in Computer Science (CERI)

Avignon, France
2020-2022

M.Sc in ICT for Internet and Multimedia (ERASMUS+ Mobility)

► University of Padova, Italy

Padova, Italy
2019-2020

Engineer in Telecommunications

► Higher School of Communication of Tunis, University of Carthage

Tunis, Tunisia
2017-2020

Paper & Certifications

- ❖ HAPe: Optimizing Customer Relation by Automatic Task Distribution using Constrained Optimization and Natural Language Processing (**CIRED 2023 - Enedis**)
Août. 2023
- ❖ Mathematics For Machine Learning Specialization (**Coursera**)
Nov. 2020
- ❖ Neural Network and Deep Learning (**Coursera**)
Nov. 2019

Personal Info



Montpellier, France



(+33) 7 49 82 75 89



abdoulaye.sayoutisouleymane@gmail.com



<https://github.com/Abdoulaye-Sayouti>



<https://www.linkedin.com/in/sayouti-souleymane-abdoulaye/>

Industries

- Telecommunications.
- Electricity.
- Building information modeling (BIM).
- Dental.
- Semiconductor.

Skills

- **Programming Languages**
Python, C, C++, Java, JavaScript
- **Machine & Deep Learning**
Scikit-Learn, PyTorch, TensorFlow, Transformers, Vertex AI (GCP)
- **NLP**
LLM, RAG, LlamaIndex, LangChain, LangGraph, Prompt Engineering
- **Data Processing**
 - **3D:** Open3D, PyTorch3D, MeshLab, IfcOpenShell
 - **General:** NumPy, Pandas, Hadoop, Talend
- **Databases**
MySQL, PostgreSQL, Elasticsearch
- **Data Visualization**
Tableau, Kibana, Dash
- **Cloud, DevOps & MLOps**
Docker, GitLab, Jenkins, MLflow, Cloud Storage & Dataflow (GCP)
- **Web/API Development**
Flask, ReactJS, FastAPI

Languages

- ❖ French
- ❖ English